

Mrityunjai Rathore

+91 7014322325 ✉ mrityunjairathore19n@gmail.com  linkedin.com/in/mrityunjai-rathore
 github.com/Mrityunjai123

Professional Summary

Junior Computer Science Engineering student at Bennett University specializing in machine learning and data analytics. Experienced in developing robust, state-of-the-art generative models and neural network architectures. Proven ability to lead and deliver complex projects, including a successful internship at the Defence Research and Development Organization (DRDO) where I engineered an Enhanced Offline RAG System for secure document retrieval. Passionate about applying analytical skills to solve real-world problems, particularly in humanitarian and healthcare sectors.

Technical Skills

- **RLHF & Alignment:** Data Annotation, Intent Labeling, RAG (Retrieval-Augmented Generation), Hallucination Mitigation, Prompt Engineering, Model Evaluation, Safety Guardrails.
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Scikit-learn, FinBERT, OpenCV, Computer Vision.
- **Data & Tools:** Pandas, NumPy, ChromaDB (Vector Store), SQL, Git, Docker, Label Studio/Annotation Tools.

Professional Experience

Square Group LLC

Chantilly, VA

AI & Data Science Intern (Remote)

Aug 2025 – Present

- **Architectural Segmentation (New Jersey Township Project):** Spearheaded the creation of a segmentation dataset for architectural drawings using data annotation tools. Developed an end-to-end ML pipeline for the automated detection and labeling of architectural elements, facilitating accurate structure analysis for the township project.
- **AI Safety & Guardrails:** Engineered safety guardrails using Regex patterns to post-process LLM outputs, preventing hallucinated SQL commands and ensuring strict adherence to data security schemas.
- **Model Alignment (RAG):** Architected a Retrieval-Augmented Generation (RAG) pipeline using LangChain and ChromaDB to ground LLM responses in factual database schemas, effectively eliminating model hallucinations.

Defence Research and Development Organization (DRDO)

Jodhpur, India

AI Research Intern

June 2025 – July 2025

- **Offline RAG Evaluation:** Developed a secure, offline RAG system for sensitive document retrieval, prioritizing factual accuracy over generation creativity.
- **Anti-Hallucination Optimization:** Achieved a **96% accuracy rate** in factual QA by implementing strict context-verification logic to prevent model hallucinations.
- **Performance Engineering:** Integrated GPU acceleration to reduce query response latency by 9.5x, enabling real-time evaluation of large document corpora.

Key Projects

1. Enhanced Offline RAG System | (Python, PyTorch, RAG, LangChain, Streamlit)

June 2025 – July 2025

 - Developed a comprehensive offline RAG system for intelligent document processing and question answering in secure environments.
 - Implemented a modular architecture featuring a document processing pipeline, transformer-based embeddings, a FAISS vector store, and a local LLM for response generation.
 - Created a user-friendly interface with Streamlit, providing three interaction modes (Q&A, teaching, summarization) and ensuring accessibility for non-technical users.
2. 3D U-Net Tumor Segmentation w/Spatial Attention | (TensorFlow, Medical Imaging)

 [GitHub](#) Developed

 - 3D U-Net model with spatial attention layers in Conv3D blocks, achieving 85% mIoU on the BraTS dataset.
 - Trained using a custom Dice-Cross entropy loss, resulting in 18% fewer false positives compared to the baseline U-Net.
3. Data-Driven Analysis of the Russia-Ukraine Conflict | (Python, Scikit-learn, ARIMA, Prophet)

Ongoing

 - Conducted a comprehensive, multidimensional analysis of the humanitarian, economic, and environmental impacts of the conflict using high-resolution datasets.
 - Applied advanced statistical models (ARIMA, SARIMA, Prophet), Bayesian methods, and spatial econometrics to reveal critical patterns influencing humanitarian outcomes.
 - Provided actionable insights for international humanitarian response and policy planning.
4. Text-to-Image Generation and 2D-to-3D Conversion | (PyTorch, Diffusion Models, Hugging Face)

 [GitHub](#)

 - Implemented Stable Diffusion models, achieving a high inception score of 9.1 for realistic image generation.
 - Utilized Hugging Face frameworks to efficiently convert 2D images into 3D models, exploring advanced generative AI applications.

EDUCATION

Sr. no.	Degree	University	Percentage	Year
1.	B.Tech in Computer Software Engineering	Bennett University Greater Noida, Uttar Pradesh	8.0/10	2022 –2026

EXTRA CURRICULAR

- Certificate Course in Share Data through the Art of Visualization by Google.
- Certificate Course in Google Data Analytics.